Appendix: Supplementary Evidence for JDCNet

Bo Ma[1*]

[1*]Auckland University of Technology.

Appendix A Supplementary Evidence and Interpretation

This appendix documents how the patient-level comparisons were computed and read. All entries are support for, or constraints on, the main-text findings rather than independent clinical validation.

A.1 How to Read the Evidence

Each comparison is paired: for each held-out fold and seed, the supervised X-ray model and the tested model are evaluated on the same patients, and the tables report the tested model's balanced accuracy (BA) minus the matched supervised BA. BA gives equal weight to sensitivity and specificity, which matters for the imbalanced 510-patient cohort. The decision rule (mean $\Delta\text{BA} \geq +0.03$ and lower 95% bootstrap CI bound > 0) is a descriptive screen, not a confirmatory test; because the JDCNet configurations were explored adaptively on the same cohort (main text, Section 3.7), multiplicity-adjusted p-values and unselected averages are reported alongside it. CT-only reference rows indicate whether the CT view carries class signal for the same patients; transfer rows indicate whether a training objective moves the X-ray student.

A.2 Directly Computed Tests for All JDCNet Configurations

The per fold/seed test-partition records of all 16 configurations are archived, so every test can be computed directly from the 15 paired cells (Table A1). The paired t-test, Wilcoxon signed-rank test, bootstrap test, and exact sign test lead to the same reading: a few configurations have unadjusted $p < 0.05$, none remains significant after Benjamini–Hochberg adjustment over the 16 configurations, and the per-cell average over all 16 is positive. The mean change in specificity is positive for all 16 configurations, whereas the mean change in sensitivity is negative for 12.

Table A1 Tests computed directly from the archived per fold/seed records of the 16 JDCNet configurations (510-patient cohort; n=15 paired cells; differences against the batch-512 supervised baseline). Stage: I, initial; A, extension A; B, extension B. p_t: paired t-test; p_W: Wilcoxon signed-rank; p_b: bootstrap (shift method, 10,000 resamples); p_s: exact sign test; p_{BH}: Benjamini–Hochberg-adjusted p_t. ΔSens, ΔSpec, ΔAUC: mean paired differences in sensitivity, specificity, and AUC.

View	τ	λ	Target	Stage	ΔBA	p_t	p_W	p_b	p_s	p_{BH}	ΔSens	ΔSpec	ΔAUC
3-slice	0.70	1.0	soft	B	+0.0345	0.014	0.022	0.003	0.30	0.22	−0.022	+0.091	+0.039
Mid	0.80	1.5	hard	A	+0.0329	0.028	0.035	0.010	0.18	0.22	+0.013	+0.053	+0.018
Mid	0.70	1.0	hard	I	+0.0298	0.079	0.095	0.050	0.30	0.22	−0.001	+0.061	+0.033
Mid	0.70	1.5	hard	A	+0.0296	0.061	0.14	0.032	0.61	0.22	−0.005	+0.064	+0.006
Mid	0.70	1.0	soft	B	+0.0264	0.082	0.14	0.051	0.30	0.22	+0.016	+0.036	+0.022
3-slice	0.80	1.0	soft	B	+0.0258	0.11	0.14	0.082	0.30	0.23	−0.021	+0.073	+0.026
3-slice	0.80	1.0	hard	I	+0.0247	0.082	0.17	0.050	0.30	0.22	−0.037	+0.086	+0.027
3-slice	0.70	0.5	hard	I	+0.0239	0.11	0.14	0.087	0.30	0.23	−0.005	+0.053	+0.018
Mid	0.80	1.0	soft	B	+0.0231	0.19	0.23	0.15	0.61	0.33	−0.044	+0.091	+0.018
3-slice	0.80	1.5	hard	A	+0.0181	0.25	0.25	0.22	0.61	0.36	−0.018	+0.054	+0.038
3-slice	0.80	0.5	hard	I	+0.0151	0.40	0.30	0.37	0.61	0.54	+0.016	+0.014	+0.018
Mid	0.80	0.5	hard	I	+0.0144	0.22	0.23	0.18	0.30	0.34	+0.016	+0.013	+0.014
Mid	0.80	1.0	hard	I	+0.0127	0.44	0.49	0.41	0.30	0.54	−0.019	+0.045	+0.016
3-slice	0.70	1.0	hard	I	+0.0066	0.63	0.76	0.63	1.00	0.72	−0.052	+0.065	+0.012
Mid	0.70	0.5	hard	I	+0.0005	0.97	0.98	0.97	1.00	0.97	−0.061	+0.062	+0.006
3-slice	0.70	1.5	hard	A	−0.0031	0.84	0.93	0.83	0.61	0.90	−0.077	+0.071	0.000
Average of the 8 initial					+0.0160	0.11	0.28	0.078	1.00	–	–	–	–
Average of all 16					+0.0197	0.060	0.026	0.032	0.007	–	–	–	–

A.3 Cohort Construction and Leakage Audit

The 510-patient cohort was assembled with the following rules. (i) Patients are taken from the BIMCV-COVID19+ release (label COVID-19) and the BIMCV-COVID19− release (label non-COVID); every patient for whom at least one chest CT volume and at least one chest radiograph could be retrieved is included. No acquisition-date window was imposed between the two studies. (ii) Each patient contributes exactly one radiograph and one CT volume: the largest CT file (typically the full-volume acquisition) and one radiograph, chosen deterministically as the first after sorting by file path. All selected radiographs are anteroposterior. (iii) Labels are patient-level and inherited from the release. (iv) Patients are assigned to five folds by a seeded random permutation within each class. For fold k, fold k is the test partition, fold $(k{+}1) \bmod 5$ the validation partition, and the remaining three folds the training partition. (v) Because each patient has one row, no patient's radiograph or CT appears in more than one partition of a split, and all methods are evaluated on identical test patients within each fold and seed. (vi) CT teachers, supervised baselines, and JDCNet students are trained only on the training partition of each fold and seed, with checkpoint selection on the validation partition; test partitions are used only for the final evaluation. (vii) Inputs are scaled to $[0, 1]$ without dataset-level intensity statistics, so no normalization statistics are shared across partitions.

Rule (i) has two consequences for interpretation. The CT and radiograph of a patient may come from different visits, which weakens the pairing; and because the two classes come from two releases, a classifier may exploit release-specific acquisition cues (main text, Section 4.10). Figure A1 summarizes the construction.

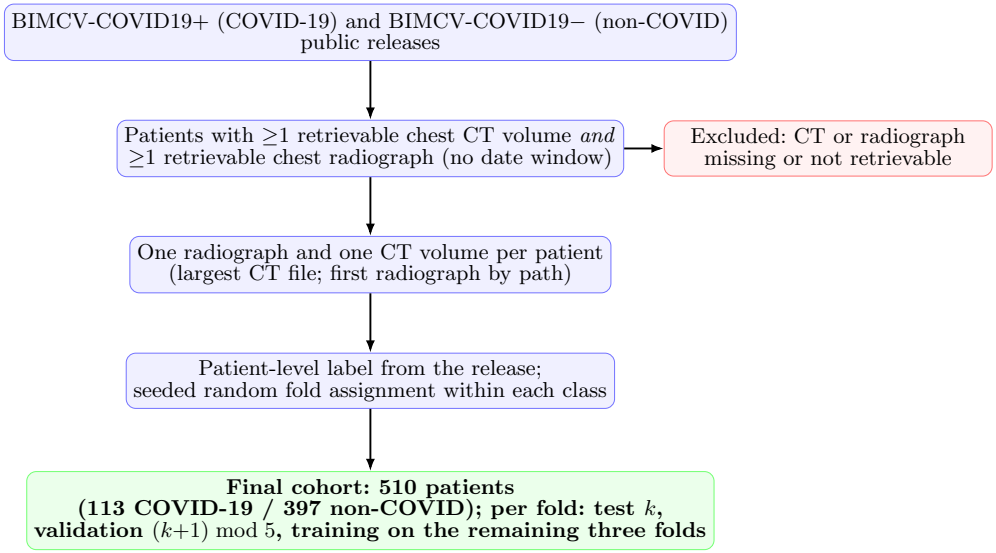


Fig. A1 Construction of the 510-patient same-patient paired BIMCV cohort, following rules (i)–(iv). Teachers, baselines, and students are trained within the training partition of each fold and seed (rule (vi)).

A.4 Comparator Interpretation

Table A2 reads the CT-only references and the transfer rows together. The CT-only references show that the mid-slice and 3-slice CT views carry class signal for the same patients. The transfer rows show that, among the tested channels, only the JDCNet configurations are mostly positive, and that their advantage survives a nested, batch-matched comparison within this cohort.

A.5 Full Contrastive-Alignment Sweep

Table A3 reports all four contrastive-alignment configurations. Two teacher views (mid-slice and 3-slice, the views whose CT-only references exceed the baseline) and two InfoNCE temperatures ($\tau_{\text{NCE}} \in \{0.07, 0.20\}$) were evaluated with the same folds, seeds, ResNet-18 student, and weighted sampler as the JDCNet sweep. The two-stage protocol (1) trains a paired CT/X-ray encoder pair with projection heads using symmetric InfoNCE on patient-paired batches for 100 epochs, then (2) discards the CT branch, replaces the X-ray head with a linear classifier, and fine-tunes for 50 epochs with weighted cross-entropy on the same labeled training partition. No configuration shows a gain; the best (3-slice, τ_{NCE}=0.20) yields +0.008 with 7 positive, 1 tied, and 7 negative cells, and both mid-slice configurations are negative on the point estimate.

A.6 CT View Comparison on the 226-Patient Subset

Before the 510-patient experiments, the four CT views (mid-slice, 3-slice stack, axial mean projection, DRR) were compared on a class-balanced 226-patient BIMCV subset with five-fold patient-level cross-validation and seeds 42–44 (Table A4). Each view was

Table A2 Comparator interpretation. Unless marked otherwise, rows are paired differences against the supervised X-ray baseline on the 510-patient cohort.

Family	Representative result	Reading
CT-only reference	Mid $+0.045$ $[+0.019, +0.070]$; 3-slice $+0.051$ $[+0.025, +0.079]$	CT carries signal
JDCNet	15/16 configurations positive; nested, batch-matched comparison $+0.037$ $[+0.016, +0.058]$	Small within-cohort gain
Gated logit KD	Mid -0.022 $[-0.055, +0.010]$; 3-slice -0.026; mean proj. -0.014; DRR -0.064 $[-0.095, -0.034]$	No gain
Plain logit KD	3-slice -0.036 $[-0.062, -0.013]$; other views -0.016 to -0.001, CIs include zero	No gain
Contrastive alignment	Best of 4: $+0.008$ $[-0.021, +0.038]$	No gain
BiomedCLIP (226 subset)	$+0.002$ $[-0.048, +0.050]$ vs. ResNet-18 on the same subset	No gain
Gated logit KD (earlier balanced-subset scan)	Best of eight configurations $+0.034$, CI crosses zero	Not reproduced at 510

Table A3 Contrastive alignment (InfoNCE pretraining followed by supervised fine-tuning) on the 510-patient cohort ($n = 15$ paired fold/seed cells per row).

Teacher view	τ_{NCE}	BA mean [95% CI]	ΔBA [95% CI]	$+/0/-$
3-slice stack	0.20	0.633 [0.610, 0.656]	$+0.008$ $[-0.021, +0.038]$	7/1/7
3-slice stack	0.07	0.627 [0.594, 0.657]	$+0.003$ $[-0.023, +0.030]$	7/0/8
Mid-slice	0.20	0.620 [0.594, 0.643]	-0.005 $[-0.030, +0.020]$	5/0/10
Mid-slice	0.07	0.616 [0.590, 0.642]	-0.008 $[-0.038, +0.027]$	5/0/10

used for a CT-only reference, plain logit KD, and gated logit KD (T=4, α=0.6, gate threshold 0.50 with the teacher-correctness filter of Eq. (5)). On this subset only the mean-projection CT-only reference clearly exceeded the X-ray baseline, and no logit-KD configuration showed a gain for any view. On the 510-patient cohort the mid-slice and 3-slice references exceeded the baseline while the mean projection did not, and the logit-KD comparators remained negative or neutral for all four views.

A.7 Earlier Gated Logit-KD Scan on a Balanced Subset

Before the CT-view comparison of Table A4, a scan of gated logit KD over eight combinations of $T \in \{2, 4, 8\}$ and gate threshold $\in \{0.50, 0.55, 0.60\}$ (T=4 with threshold 0.55 was not run; five folds $\times$ three seeds per configuration) was run on a class-balanced BIMCV subset of similar size, with a different split, teacher run, and training configuration, and against the baseline of that run. Its best configuration (T=4, threshold 0.50) reached $\Delta\text{BA} = +0.034$ with a CI crossing zero ($[-0.004, +0.073]$). Because the split and training configuration differ, this value is not directly comparable with the gated-KD rows of Table A4, which use the same T and threshold but are negative or neutral for every view. With eight configurations scanned, this best value is itself a selected estimate, and on the 510-patient cohort the gated-logit configurations were

Table A4 CT view comparison on the class-balanced 226-patient BIMCV subset (five folds × seeds 42–44, n=15 cells per row). ΔBA is the paired difference against the supervised X-ray baseline trained on the same subset.

Teacher view	Method	ΔBA [95% CI]	+/0/−
Mid-slice	CT-only ref.	+0.011 [−0.024, +0.046]	8/0/7
	Plain KD	−0.022 [−0.062, +0.021]	6/0/9
	Gated KD	−0.014 [−0.053, +0.023]	8/0/7
3-slice stack	CT-only ref.	+0.019 [−0.013, +0.058]	8/3/5
	Plain KD	−0.030 [−0.083, +0.016]	7/0/8
	Gated KD	−0.045 [−0.092, +0.000]	5/0/10
Mean projection	CT-only ref.	+0.045 [+0.008, +0.081]	9/1/5
	Plain KD	−0.031 [−0.082, +0.019]	5/1/9
	Gated KD	−0.015 [−0.058, +0.027]	6/2/7
DRR	CT-only ref.	+0.023 [−0.023, +0.066]	9/0/6
	Plain KD	−0.043 [−0.095, +0.014]	4/1/10
	Gated KD	−0.005 [−0.048, +0.039]	8/0/7

negative or neutral. The scan illustrates the selection and small-sample effects that the main text addresses for JDCNet.

A.8 Earlier Gate-Temperature Reruns on a Different Host

Before the pre-registered experiment of Section 4.7, the two selected configurations were rerun on a different host with the supervised baseline and JDCNet trained with the same batch size (128) and schedule, varying the gate temperature $T_g \in \{0.5, 1.0, 2.0\}$. For a two-class teacher this is equivalent to changing the raw-confidence threshold (main text, Eq. (4)); the equivalent raw thresholds are listed in Table A5. The rerun used a separate five-fold split of the BIMCV paired pool, restricted to the 466 patients for whom a CT teacher view could be rendered on the rerun host, a single seed (n=5 folds per row), and reports BA on the validation partition, which was also used for checkpoint selection; values are therefore optimistic for both arms and not comparable with the test-fold values of the main sweep. Both configurations are positive at every temperature, and within-configuration changes are at most 0.025. The 3-slice soft-KL intervals at T_g=1.0 and 2.0 include zero. An earlier rerun of the same design on a class-balanced BIMCV subset (216 patients with a rendered CT teacher view) with three seeds did not reproduce a gain (3-slice soft-KL about −0.09 with a CI excluding zero; mid-slice hard about −0.02 with a CI including zero; within-configuration changes across temperatures at most 0.013). The per-run records of both reruns were later recovered from a mirror of that host; JDCNet test-partition metrics exist, but the supervised baselines of these reruns were evaluated only on the validation partition, so the paired comparison is available only on that partition. Both reruns are superseded by the pre-registered experiment. The reruns were prepared on a different host from the main sweep. On that host, CT volumes were not available for some positive patients; for those patients the preprocessing approximated the 3-slice and mean-projection views by replicating a previously extracted mid-slice image, and the number of affected patients was not recorded. The 3-slice teacher of both reruns is therefore partly a mid-slice teacher, which further limits their comparability with the main sweep, where every view was rendered from the CT volume.

Table A5 Batch-matched gate-temperature rerun on a 466-patient split (single seed, n=5 folds per row; validation-partition BA). τ': raw-confidence threshold equivalent to gating at τ with temperature T_g. ΔBA is the paired difference against the batch-matched supervised baseline (validation BA 0.664 for the 3-slice row set and 0.604 for the mid-slice row set, which use different student manifests); the last column is the change relative to T_g=1.0.

Configuration	T_g	τ'	BA	ΔBA [95% CI]	vs. T_g=1.0
3-slice soft-KL, τ=0.70	0.5	0.60	0.711	+0.047 [+0.028, +0.064]	+0.025
3-slice soft-KL, τ=0.70	1.0	0.70	0.686	+0.022 [−0.011, +0.068]	0.000
3-slice soft-KL, τ=0.70	2.0	0.84	0.691	+0.027 [−0.001, +0.061]	+0.005
Mid hard, τ=0.80	0.5	0.67	0.680	+0.076 [+0.035, +0.117]	+0.007
Mid hard, τ=0.80	1.0	0.80	0.673	+0.069 [+0.034, +0.106]	0.000
Mid hard, τ=0.80	2.0	0.94	0.659	+0.054 [+0.024, +0.088]	−0.014

A.9 Pre-Registered Supplementary Experiment

The analysis plan for this experiment was written and committed to the code repository before any run. On the original 510-patient splits and with the original training code, 16 supervised restarts were trained in each of the 15 fold/seed cells at batch size 128 (training seeds $s + 1000r$ for cell seed s and restart $r = 0, \ldots, 15$; 240 runs); everything else matched the original supervised configuration. Every checkpoint of the original sweep and of the restarts was then re-run to store per-patient probabilities on the test partition and on the external set; the recomputed test BA matched the recorded value for 613 of the 615 runs with recorded test metrics and differed by 0.006 and 0.022 BA (one patient) for the other two, one initial-grid JDCNet run and one supervised restart. All cell-level statistics use the recorded values; the patient-level and external analyses use the stored predictions. Table A6 lists the nested and summary comparisons, and Table A7 lists the batch-matched comparison of every configuration (pre-specified analysis 1) with its patient-level bootstrap (analysis 3). Against r=0, all 16 configurations have a positive mean difference, and none remains significant after Holm adjustment at either level. The pre-specified external comparison (analysis 4) is reported in Section 4.10: McNemar tests per checkpoint pair gave $p < 0.05$ in 2 of 15 pairs against the batch-512 students, both favoring the supervised model, and in none of 15 against the batch-128 students.

Two features of the nested comparison are worth noting. First, validation-based selection chose 11 different JDCNet configurations across the 15 cells, and never the configuration with the best test-partition mean (3-slice soft-KL), so the advantage is a property of the configuration family rather than of one configuration. Second, the selected supervised restarts were spread over 11 different restart indices, and their mean validation BA (0.731) was close to that of the selected JDCNet runs (0.745), so the supervised baseline was not handicapped in the selection step.

Table A6 Pre-registered supplementary comparisons on the 510-patient splits (batch-matched supervised restarts). Cell-level: paired over 15 fold/seed cells (bootstrap CI, paired t-test). Patient-level: paired bootstrap over 510 patients with out-of-fold predictions pooled per seed (10,000 resamples).

Comparison	Cell-level ΔBA [95% CI]	p	Patient-level ΔBA [95% CI]	p
Supervised batch 128 (r=0) − batch 512	−0.005 [−0.032, +0.022]	0.75	−0.004 [−0.038, +0.029]	0.82
Supervised best of 16 − supervised r=0	+0.013 [−0.011, +0.038]	0.33	–	–
JDCNet best of 16 − supervised best of 16 (primary)	+0.037 [+0.016, +0.058]	0.005	+0.037 [+0.007, +0.068]	0.017
JDCNet best of 8 − supervised best of 8	+0.029 [+0.007, +0.049]	0.022	–	–
JDCNet best of 16 − supervised batch 512	+0.046 [+0.021, +0.072]	0.004	+0.046 [+0.010, +0.082]	0.012
3-slice soft-KL − supervised r=0	+0.039 [+0.002, +0.074]	0.063	+0.039 [+0.003, +0.077]	0.036
Mid hard − supervised r=0	+0.038 [+0.009, +0.069]	0.035	+0.038 [+0.001, +0.073]	0.045
Average of 16 configurations − mean of 16 restarts	+0.019 [+0.006, +0.031]	0.016	–	–

Table A7 Pre-specified batch-matched comparison of every JDCNet configuration against the first batch-128 supervised restart (r=0) of the same cell (510-patient cohort; rows in the order of Table A1, where the exploration stage of each configuration is listed). Cell level: mean paired difference over the 15 fold/seed cells, bootstrap CI, and paired t-test p_t. Patient level: paired bootstrap over the 510 patients with out-of-fold predictions pooled per seed (10,000 resamples) and two-sided bootstrap p_b. p_{H}: Holm-adjusted over the 16 configurations.

				Cell level					Patient level		
View	τ	λ	Target	+/0/−	ΔBA	95% CI	p_t	p_{H}	95% CI	p_b	p_{H}
3-slice	0.70	1.0	soft	11/0/4	+0.039	[+0.002, +0.074]	0.063	0.95	[+0.003, +0.077]	0.036	0.54
Mid	0.80	1.5	hard	9/0/6	+0.038	[+0.009, +0.069]	0.035	0.56	[+0.001, +0.073]	0.045	0.63
Mid	0.70	1.0	hard	9/0/6	+0.034	[+0.000, +0.071]	0.086	1.00	[−0.001, +0.071]	0.061	0.72
Mid	0.70	1.5	hard	9/1/5	+0.034	[+0.004, +0.069]	0.064	0.95	[+0.005, +0.064]	0.021	0.33
Mid	0.70	1.0	soft	9/0/6	+0.031	[−0.000, +0.064]	0.091	1.00	[−0.000, +0.062]	0.050	0.66
3-slice	0.80	1.0	soft	10/0/5	+0.030	[−0.005, +0.065]	0.12	1.00	[−0.002, +0.063]	0.065	0.72
3-slice	0.80	1.0	hard	9/1/5	+0.029	[−0.006, +0.064]	0.14	1.00	[−0.002, +0.062]	0.069	0.72
3-slice	0.70	0.5	hard	10/0/5	+0.029	[+0.000, +0.057]	0.077	1.00	[−0.001, +0.060]	0.060	0.72
Mid	0.80	1.0	soft	10/0/5	+0.028	[−0.011, +0.065]	0.19	1.00	[−0.008, +0.065]	0.13	1.00
3-slice	0.80	1.5	hard	9/0/6	+0.023	[−0.016, +0.064]	0.30	1.00	[−0.017, +0.061]	0.26	1.00
3-slice	0.80	0.5	hard	11/0/4	+0.020	[−0.003, +0.041]	0.11	1.00	[−0.012, +0.052]	0.20	1.00
Mid	0.80	0.5	hard	10/0/5	+0.019	[−0.011, +0.051]	0.26	1.00	[−0.011, +0.051]	0.21	1.00
Mid	0.80	1.0	hard	9/0/6	+0.017	[−0.016, +0.049]	0.33	1.00	[−0.015, +0.050]	0.29	1.00
3-slice	0.70	1.0	hard	9/1/5	+0.011	[−0.012, +0.035]	0.39	1.00	[−0.024, +0.046]	0.52	1.00
Mid	0.70	0.5	hard	9/0/6	+0.005	[−0.026, +0.034]	0.75	1.00	[−0.030, +0.040]	0.77	1.00
3-slice	0.70	1.5	hard	8/0/7	+0.002	[−0.035, +0.039]	0.94	1.00	[−0.036, +0.037]	0.95	1.00
Average of all 16				10/0/5	+0.024	[−0.000, +0.051]	0.092	–	–	–	–